SYSTEM OVERVIEW · VOICE AI · MULTI-AGENT

Delivery Desk

A voice agent that answers “where is my order?” calls. It checks who is calling, explains what happened to the package, fixes the problem, and emails a confirmation. All in one natural conversation that the caller can interrupt at any time.

FORMAT	STATUS	AGENTS	AUTOMATED TESTS
Live voice call	**Working prototype**	**3 specialists**	**52 passing**

A call, start to finish

Three specialist agents share one conversation. The caller hears a single voice, and the hand-offs between agents are silent.

1

Greeter

Welcomes the caller and finds out which order they mean, by order number or by the phone number on the account. It reads the number back in small groups, because speech recognition mishears digits.

`Hand-off to Order Status once the caller confirms the number.`

2

Order Status

Asks for the last four digits of the phone number on the order. Only after those match does it share anything. It then explains the status in plain speech: why a package is late, where it is, and when it should arrive.

`look up by order number` `look up by phone` `verify caller` `tracking history`

`Hand-off to Resolution when the caller wants something changed.`

3

Resolution

Reschedules a delivery, requests a refund, opens a support ticket, or passes the caller to a person. It says exactly what it is about to do and waits for a clear “yes”. After the action, a confirmation email is sent and the caller is told it is on its way. If the caller asks about the order again, the call goes back to Order Status.

`reschedule delivery` `request refund` `open ticket` `escalate to a person` `confirmation email`

What it sounds like

CALLER	Hi, my espresso machine is really late. The order number is O R D 4 8 2 1 3.
AGENT	Oh no, sorry to hear that. Just to confirm, that's O R D, 4 8 2 1 3, right?
CALLER	Yes. Zero one four two.
AGENT	Got it, Maya. Your espresso machine hit a weather hold at the Denver hub because of a winter storm, and it's still paused there as of this afternoon. Would you like to... CALLER CUTS IN
CALLER	Wait, can you just move the delivery to Friday?
AGENT	Sure. I'd reschedule it to this Friday, October 30th. Does that sound right?
CALLER	Yes, go ahead.
AGENT	All set. It's moved to Friday, and a confirmation email is on its way. Anything else I can help with?

An illustrative exchange based on test calls. The customer and order are fabricated demo data.

Why it feels like talking to a person

It waits when you pause

Most voice bots answer after a fixed half second of silence, so they cut in at "my order number is...". This one reads the sentence as it arrives and judges whether it is finished.

You can interrupt it

When the caller starts talking, the agent stops speaking straight away and answers the new question. It remembers only what the caller actually heard.

It replies quickly

Once the caller has finished a sentence, the agent waits 0.3 seconds. In testing, its first words arrive about one to two seconds after the caller stops, and speech starts within about 0.2 seconds of the text being ready.

It talks like a person

Short, varied replies such as "Got it." and "Oh no, sorry about that." It says one brief "one moment" before looking something up, and never announces a transfer.

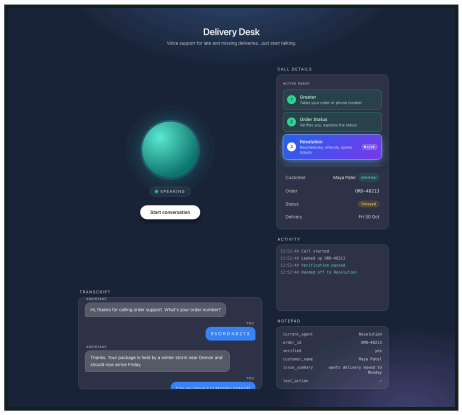
What it will not do

These rules live in the code that runs the tools, so the language model cannot talk its way around them.

Share order details before verification	Every order lookup, tracking history and action is blocked until the last four phone digits match. After three wrong tries it stops and offers a person.
Repeat the verification digits	The agent never reads those four digits back, so a bystander on the line learns nothing.
Act without a clear yes	Rescheduling, refunds and tickets are described out loud first and only run after the caller agrees.
Do things twice	If the caller interrupts while a refund or email is in progress, it still completes once. No duplicate tickets, no duplicate emails.
Invent the confirmation email	The email is written by code from the saved record of what was done, never from text the model produced.
Leave the caller stuck	If the model or a tool fails, the agent apologises and asks the caller to repeat. A caller can ask for a person at any point, even before verifying.

What the operator sees

A live panel beside the call shows which agent is talking, whether the caller is verified, the order and its status, a running log of every action taken, and the shared call notes. It makes the system easy to demo and easy to debug.



The browser client during a sample call. The agent list, customer, activity log and notepad update in place as the call moves along.

Scope

This is a working prototype on demo data. Here is what is finished, what a live deployment would add, and what is not covered.

BUILT AND DEMONSTRATED

- Natural two-way voice conversation in the browser, with interruptions
- Three cooperating agents with silent hand-offs
- Caller verification enforced in code
- Order status, delay reasons and tracking history
- Reschedule, refund, support ticket and escalation
- Confirmation email after every action
- Live operator panel
- 52 automated tests, plus scripted end-to-end calls

NEEDED TO GO LIVE

- Connect your real order system and carrier tracking in place of the demo database
- A phone number, so customers can dial in instead of using a web page
- Your refund and rescheduling rules, such as limits and approvals
- A live hand-off to your support team, with the call summary passed along
- Privacy review: call recording consent, data retention, personal data handling
- Load testing and monitoring at your expected call volume

NOT COVERED YET

- Languages other than English
- Outbound calls and proactive delay notices
- Payment handling
- Compliance certification of any kind
- Call analytics dashboards

Where else this pattern fits

The structure is a small team of specialist agents, each with its own tools, a shared memory of the call, and rules in code that guard every sensitive action. That suits any phone line where someone has to be identified, a question answered and a task completed. Examples are appointment booking for clinics and salons, front desk and reservations, internal IT helpdesks, and first-line lead qualification.

Technology

ROLE	TOOL	WHY
Real-time voice	`LiveKit Agents`	Live audio, interruptions and turn-taking
Speech to text	`Deepgram nova-3`	Streams the transcript while the caller is still talking
End of turn	`Silero VAD + semantic turn model`	Tells a pause from a finished sentence
Text to speech	`Cartesia sonic-3`	Natural voice, first audio in about 0.2 seconds
Agent logic	`LangGraph, LangChain`	Agents, tools and call state as an explicit graph. The language model behind it can be swapped.
Backend	`Python 3.11, FastAPI, SQLite`	Token service and the demo order database
Email	`Gmail SMTP`	Confirmation messages
Client	`Single-file web page`	Voice call plus the live operator panel

How a project would run

PHASE 1

Scope the line

Pick the calls worth automating, write down the rules, and decide what stays with people.

PHASE 2

Build on your data

Adapt the agents and prompts to your business, and test them against your real scenarios.

PHASE 3

Connect live systems

Plug into your order system, carriers, email and phone provider.

PHASE 4

Pilot and hand over

Run on a slice of real calls, tune from what people actually say, then document and hand over.

All customers, orders and emails shown here are fabricated for the demo. No real personal data is used.